Born January 1, 1990
Example citizen

jane.doe@example.org
https://example.org
https://github.com/example

Researcher in Statistics

Example Institute
1 Example Street
00000 Example City

Experience

2011–	Researcher, Example Institute
2008–2011	Postdoc, Example University
supervisor	Dr. Example

Education

2015	Habilitation in Mathematics
Title	*Sparse methods for complex data*
2003	PhD in Applied Mathematics

Grants

On going Projects

2023–2027	DISCERN – Discovering the causes of a poorly understood cancer
Partners	20 partners
Support	Horizon Europe
Description	A five-year consortium project combining statistical genomics, imaging, and epidemiological cohort data to identify early biomarkers of a poorly understood cancer subtype. My group leads the statistical modelling work package, developing sparse graphical models for multi-omics integration across the twenty partner sites.

Past

2019–2023	EcoNet – Advanced statistical modelling of ecological networks
Support	French National Research Agency (ANR)
2017–2019	SEARS – Sampling strategies and network analysis
Leader	Mathieu Thomas

Other grants

2010–2012	Older grant, listed only in the long version
Support	French National Research Agency (ANR)

Activities

Scientific committees

	Councils
since 2022	Member of the Foundation Council
2020–2024	PhD Committee (full list)
role	Reviewer for 12 PhD theses

Students

PhD students

2021–2024 — Jane Roe – Sparse statistical methods for large-scale multi-omics data
Co-supervisor — Prof. Example
Description — Developed sparse graphical model estimators for integrating heterogeneous omics data sources, with applications to the DISCERN consortium. Defended in 2024 ; now a postdoctoral researcher at Example University.

Teaching

Approximately 2000 hours of teaching. Lecturer at Example University from 2015 to 2024.

2020–24 — Statistics in Action with R
Msc — Probabilistic models, data analysis, R programming

2017 — Convex Optimization (12h course)
Msc — Gradient methods, Newton method, Proximal methods

Publications

Selected Publications

[JP1] J. Doe et J. Roe. "A Toy Example for the quarto-cv-blocks Extension". In : *Journal of Reproducible Examples* 1 (2023), p. 1-10. DOI : 10.0000/example.2023.001.
[JP2] J. Doe. "Demonstrating a Fictitious Conference Paper". In : *Proceedings of the Example Conference*. 2021, p. 42-50.

Selected Talks

[CT1] J. Doe. "Reproducible CVs with Quarto". In : *Example Workshop on Reproducible Documents*. Example City, 2022.
[CT2] J. Doe et J. Roe. "A Fictitious Talk for the quarto-cv-blocks Example". In : *Proceedings of the Other Example Conference*. 2020, p. 12-20.

Research Projects

DISCERN

As part of the DISCERN project, my group develops sparse graphical model estimators for integrating multi-omics data. Given a sample covariance matrix S, we estimate a sparse precision matrix $\hat{\Theta}_\lambda$ by solving

$$\hat{\Theta}_\lambda = \arg\min_{\Theta \succ 0} \; -\log\det\Theta + \mathrm{tr}(S\Theta) + \lambda\|\Theta\|_1.$$

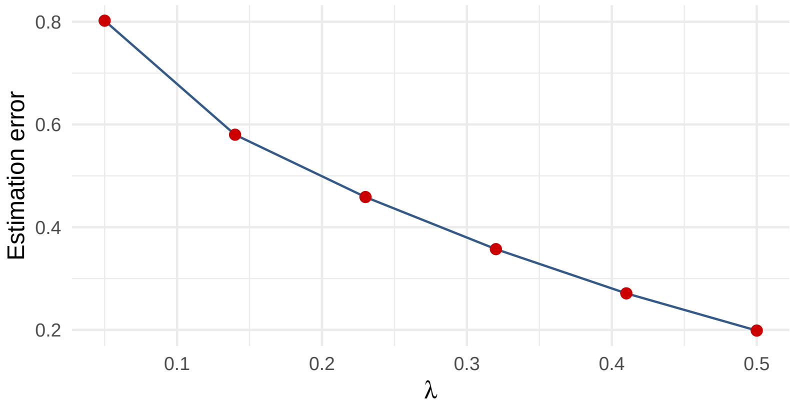

FIGURE 1 : Estimation error as a function of the sparsity penalty

Increasing the penalty λ trades estimation error for a sparser, more interpretable graph, see Figure 1.

ANOTHER PROJECT

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See Table 1.

Lambda	Error	Sparsity
0.05	0.79	12%
0.20	0.42	48%
0.50	0.19	81%

TABLE 1 : Estimation error and graph sparsity for a few penalty values